Model 1: Mixed-Integer Optimization for Child Care Capacity Planning

IEOR 4004 Project I

March 25, 2026

1 Introduction

This section formulates and solves Model 1, which minimizes the total funding required to eliminate child care capacity shortfalls across New York City's 180 zip codes while satisfying the city's policy requirement that at least two-thirds of each zip's 0–5 year old population be served in age-appropriate facilities.

Two strategies address capacity: (1) expanding existing licensed facilities, and (2) constructing new facilities. Expansion cost follows a piecewise rule distinguishing small increments (reusing existing infrastructure) from large expansions (requiring structural redesign). New facility cost is fixed per facility type, with additional equipment cost for 0–5 age group capability.

2 Formulation

2.1 Sets

$z \in \mathcal{Z}$: zip codes in NYC ($|\mathcal{Z}| = 180$).
$s \in \mathcal{S} = \{1, 2, 3\}$: facility type (Small, Medium, Large).

2.2 Core Parameters

Symbol	**Description**	**Unit**
$P_z^{(0\text{-}5)}$	Child population aged 0–5 in zip z	Children
P_z	Total child population (0–12) in zip z	Children
S_z^{curr}	Current total licensed slots in zip z	Slots
$S_z^{\text{curr,0-5}}$	Current 0–5 year capacity in zip z	Slots
$R_z^{\text{total}} = \lfloor \theta_z P_z \rfloor + 1$	Minimum slots to remove desert status	Slots
$R_z^{0\text{-}5} = \lceil \frac{2}{3} P_z^{(0\text{-}5)} \rceil$	Required 0–5 slots (city policy)	Slots
$\overline{X}_z$	Max expandable slots in zip z	Slots

2.3 Facility Types

Type	Total Slots	Max 0–5 Slots	Base Cost
Small (1)	100	50	$65,000
Medium (2)	200	100	$95,000
Large (3)	400	200	$115,000

2.4 Decision Variables

Four variable types: $x_z, x_z^{0\text{-}5}$ (expansion, continuous), y_{zs} (new facilities, integer), $u_{zs}^{0\text{-}5}$ (0–5 configuration in new facilities, continuous).

3 Cost Model and Objective

3.1 Expansion Cost (Piecewise Rule)

The expansion cost rule for facility f with current capacity n_f and expansion amount x_f is:

$$C_{\exp}(f) = \begin{cases} c_{\text{slots}}(n_f)\, x_f, & \text{if } x_f < n_f, \\ (20{,}000 + 200 n_f)\, \dfrac{x_f}{n_f}, & \text{if } x_f \geq n_f. \end{cases} \tag{1}$$

Interpretation: $x_f < n_f$ (small expansion uses existing infrastructure, low cost). $x_f \geq n_f$ (large expansion requires structural work, $20,000 base fee).

3.2 Objective Function

Zip-level aggregation with $c_z^{\exp} = \dfrac{20{,}000}{\bar{n}_z} + 200$:

$$\min \sum_z \left(c_z^{\exp} x_z + 100 x_z^{0\text{-}5} \right) + \sum_{z,s} C_s^{\text{base}} y_{zs} + 100 \sum_{z,s} u_{zs}^{0\text{-}5}. \tag{2}$$

Three components: (1) expansion renovation + 0–5 equipment, (2) new facility base cost, (3) new 0–5 equipment.

4 Constraints

$$x_z^{\text{0-5}} \le x_z \qquad \forall z \quad \text{(0–5 within expansion)} \tag{3}$$

$$x_z \le \overline{X}_z \qquad \forall z \quad \text{(expansion limit)} \tag{4}$$

$$u_{zs}^{\text{0-5}} \le K_s^{\text{0-5}} y_{zs} \qquad \forall z, s \quad \text{(0–5 config limit)} \tag{5}$$

$$S_z^{\text{curr}} + x_z + \sum_s K_s^{\text{total}} y_{zs} \ge R_z^{\text{total}} \qquad \forall z \quad \text{(desert removal)} \tag{6}$$

$$S_z^{\text{curr,0-5}} + x_z^{\text{0-5}} + \sum_s u_{zs}^{\text{0-5}} \ge R_z^{\text{0-5}} \qquad \forall z \quad \text{(0–5 policy)} \tag{7}$$

5 Model Interpretation

Model 1 jointly optimizes expansion and new construction across zips under two capacity constraints. The piecewise expansion cost captures differing marginal economics: small increments reuse infrastructure (low cost), while large increments require structural work ($20,000 base). The 0–5 configuration variables allow partial utilization of new facilities' maximum 0–5 capacity, avoiding unnecessary equipment costs.

6 Model 1 Results

6.1 Solution Status

- **Optimization Status**: 2 (OPTIMAL)
- **Optimal Total Funding**: $215,083,191
- **Solution Runtime**: 0.07 seconds

6.2 Cost Breakdown

Cost Component	Cost ($)	Share of Total
Expansion (renovation)	9,481,288	4.41%
New facility base cost	173,870,000	80.84%
0–5 equipment (expansion)	1,739,991	0.81%
0–5 equipment (new facilities)	29,991,912	13.94%
Total	**215,083,191**	**100%**

Key insight: New facility base cost dominates (80.84%), followed by 0–5 equipment for new facilities (13.94%). Equipment cost for expanded facilities is minimal, and expansion renovation itself accounts for only 4.41%. This indicates that meeting the 0–5 policy requirement is the primary cost driver.

6.3 Geographic and Facility Type Distribution

Action Pattern	**Zip Count**
Expansion only	0
New construction only	44
Both expansion and new construction	136
No action required	0
Total zips	**180**

The optimal solution employs both strategies in 136 zips and new construction alone in 44 zips. No zips use expansion alone, reflecting the cost structure where new Large facilities are more economical per slot than expanding small or medium existing facilities.

6.3.1 New Facilities by Type

Facility Type	**Count**	**Notes**
Small (100 slots)	12	Fill small residual gaps
Medium (200 slots)	5	Intermediate choice
Large (400 slots)	1,501	Dominant choice; lowest cost per slot ($287.50)

Large facilities are overwhelmingly preferred (1,501 out of 1,518 total) because the base cost per slot is significantly lower than that of small or Medium facilities.

6.4 Feasibility Verification

- Total zips in dataset: 180
- Zips in desert before optimization: 162
- Zips meeting total-slot requirement post-optimization: 180 (100%)
- Zips meeting 0–5 requirement post-optimization: 180 (100%)
- Remaining unmet capacity deficits: 0

All capacity requirements are satisfied at optimum, with no zip remaining in a desert state.